Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

New approaches to nowhere-zero flow problems

Diploma Thesis

2026
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Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

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Study Programme:	Computer Science
Field of Study:	Computer Science
Department:	Department of Computer Science
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Bratislava, 2026

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ZADANIE ZÁVEREČNEJ PRÁCE

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Študijný program: informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: New approaches to nowhere-zero flow problems
Nové prístupy k problémom o nikde-nulových tokoch

Anotácia: Táto práca nadväzuje na výsledky bakalárskej práce rovnakého autora. Novo zavedený pojem viacrozmerných Manhattanských a Čebyševovských tokov ponecháva priestor pre ďalší výskum, ako napríklad hľadanie dolných odhadov pre tokové čísla. Jedným z hlavných výsledkov spomínanej bakalárskej práce je predstavenie hypotézy tvrdiacej, že každý bezmostový graf pripúšťa (1,2)-cirkulačnú dekompozíciu, t. j. 2-cirkuláciu a 4-cirkuláciu takú, že zakaždým keď je 2-cirkulácia nulová na nejakej hrane, tak 4-cirkulácia nemôže nadobúdať 0, +1 alebo -1. To ponúka bohatý priestor na výskum vrátane rôznych zovšeobecnení, v ktorých kladieme ďalšie požiadavky na tokové hodnoty.

Cieľ: 1. Dokázať netriviálne dolné odhady pre dvojrozmerné Čebyševovské tokové číslo grafu.
2. Preskúmať možné spôsoby, ako dokázať hypotézu, že každý bezmostový graf pripúšťa (1, 2)-cirkulačnú dekompozíciu, a iné súvisiace hypotézy. Dokázať túto hypotézu pre niektoré nekonečné triedy snarkov, prípadne pre niektoré snarky, ktoré majú ďaleko od toho, aby boli zafarbiteľné (napr. s nepárnosťou 2, indexom perfektného párenia 4, ...).
3. Preskúmať ďalšie zovšeobecnenia tokov potenciálne užitočné v kontexte vyššie uvedenej hypotézy.

Vedúci: Mgr. Jozef Rajník, PhD.
Katedra: FMFI.KI - Katedra informatiky
Vedúci katedry: prof. RNDr. Martin Škoviera, PhD.

Dátum zadania: 21.11.2024

Dátum schválenia: 05.12.2024

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THESIS ASSIGNMENT

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Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: New approaches to nowhere-zero flow problems

Annotation: This work builds on the results of the bachelor thesis of the same author. The newly introduced notion of multidimensional Manhattan and Chebyshev flows still leaves some possibilities for further research like finding lower bounds on flow numbers. One of the main results of the mentioned bachelor thesis is the introduction of the conjecture asserting that each bridgeless graph admits a (1,2)-circulation decomposition, that is a 2-circulation and a 4-circulation such that whenever the 2-circulation is zero on any edge, the 4-circulations can not attain 0, +1 or -1. This offers wide possibilities of exploration including various generalisations where further requirements are posed on the flow values.

Aim:
1. Prove nontrivial lower bounds on 2-dimensional Chebyshev flow number of a graph.
2. Explore possible ways of proving the conjecture that each bridgeless graph admits a (1, 2)-circulation decomposition, and other related conjectures. Prove this conjecture for some infinite families of snarks, eventually for some snarks that are far from being colourable (e.g. with oddness 2, perfect matching index 4, …)
3. Research other generalisation of flows potentially useful in the context of the abovementioned conjecture.

Supervisor:	Mgr. Jozef Rajník, PhD.
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Acknowledgments:

Abstrakt

Kľúčové slová:

Abstract

Keywords:

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Introduction

Chapter 1

Preliminaries

The class of type II cubic graphs is widely used, so its members have special name.

Definition. A *snark* is a bridgeless cubic graph, which has girth at least 5, is cyclically 4-edge connected and is not 3-edge-colourable.

Theorem 1.1 (Menger theorem for oriented graphs)**.** [4, p. 149] *Let u, v be the vertices of an* oriented *multigraph G. Then there exist k edge-disjoint oriented uv-paths if and only if for any set $S \subseteq V(G) \setminus \{v\}$ containing u, there are at least k oriented edges from S to $V(G) \setminus S$.*

Chapter 2

Cubic graphs generation

Chapter 3

Flows

In this chapter, we aim to introduce all concepts and current knowledge about flows and related graph theory terms. Mainly, we explain Chebyshev nowhere-zero flows, which have arisen in the recent preprint coauthored by the author [5]. Moreover, we present a concept of two smaller flows as a sufficient condition for a graph to have a larger nowhere-zero flow and some Chebyshev flow, also stated in that paper.

Theorem 3.1. [6, p. 207] *There exists an NZ k-flow φ on a bridgeless graph G if and only if there also exists a NZ $\mathbb{Z}_k$-flow φ' on G. Moreover, for each φ' there exists φ satisfying $\varphi\left(\overrightarrow{d}\right) \equiv \varphi'\left(\overrightarrow{d}\right) \pmod{k}$.*

3.1 Integral nowhere-zero flows

Unless stated otherwise, we consider only unoriented simple connected graphs $G = (V, E)$. Then we can replace each unoriented edge uv with two oriented edges $\overrightarrow{uv}$, $\overrightarrow{vu}$, called *darts*. By a *dart-set* $D(S)$ of $S \subseteq V$ we mean the set of all darts with initial vertex in the set S.

Definition. Let $G = (V, E)$ be a graph and $k \geq 2$ be an integer. A *k-flow* on G is an assignment $\varphi\colon D(V) \to \{-(k-1), -(k-2), \ldots, k-2, k-1\}$ of flow values to the darts satisfying $\varphi(\overrightarrow{uv}) = -\varphi(\overrightarrow{vu})$ for any dart $\overrightarrow{uv} \in D(V)$ and

$$\sum_{\overrightarrow{d} \in D(\{v\})} \varphi\left(\overrightarrow{d}\right) = 0 \tag{3.1}$$

for each vertex v of G. The equality (3.1) is often referred to as the *flow conservation constraint*. Moreover, a *nowhere-zero k-flow* (or NZ k-flow) is a k-flow such that no flow value is zero, i.e. $\varphi\left(\overrightarrow{d}\right) \neq 0$ for any $\overrightarrow{d} \in D(V)$.

We also establish a notation $\varphi(e)$ for $e = uv \in E$, where $\varphi(e) = \max\{\varphi(\overrightarrow{uv}), \varphi(\overrightarrow{vu})\}$.

The flow conservation constraint (3.1) is generalisable to any subset of vertices. Hence, having a graph G with a bridge e, the interpretation of the flow conservation constraint for one of the components of $G - e$ results in the following observation.

Claim. If $e \in E$ is a bridge in G, then $\varphi(e) = 0$. Moreover, G allows a NZ k-flow for no natural k.

Therefore, nowhere-zero flows are examined only on bridgeless graphs.

It is not hard to see that a graph with an NZ k-flow also admits an NZ K-flow for any $K > k$. Hence it makes sense to seek for the smallest k such that a graph admits an NZ k-flow.

Definition. Let G be a bridgeless graph. A *flow number of* G is

$$\Phi(G) := \min\{k \mid \exists \text{NZ } k\text{-flow on } G\}.$$

As for many problems in the graph theory, it is sufficient to explore the problem only on the smaller subset of graphs. There exists a folklore reduction of nowhere-zero flows to 3-regular (or cubic) graphs.

Proposition 3.2. *There exists an NZ k-flow on any bridgeless graph if and only if there exists an NZ k-flow on any cubic bridgeless graph.*

Hence, by considering only cubic graphs, we still solve the universal problem. Moreover, in terms of finding a counterexample to Tutte's 5-flow conjecture, we can also avoid 3-edge-colourable ones and limit research to the cubic graphs, that are not 3-edge-colourable, as a corollary of the next proposition.

Proposition 3.3. [3, pp. 160, 161] *A cubic graph has an NZ 4-flow if and only if it is 3-edge-colourable.*

However, it turns out that this rate of complexity achieves only few values.

Theorem 3.4 (6-flow theorem)**.** [17, p. 133] *There exists an NZ 6-flow on any bridgeless graph.*

Seymour constructed this NZ 6-flow from two simpler flows from the fact, that if $a \in \{0, \pm 1\}$, $b \in \{0, \pm 1, \pm 2\}$ and not both a, b are zero, then $3a + b \in \{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$.

Lemma 3.5. [17, p. 132] *For any bridgeless graph, there exist a 2-flow φ_2 and a 3-flow φ_3 such that for each edge e, at least one value from $\varphi_2(e), \varphi_3(e)$ is non-zero.*

On the other hand, there is no known graph with $\Phi(G) = 6$, all known graphs have $\Phi(G) \leq 5$. Therefore, the conjecture about the NZ 5-flow by Tutte still holds unresolved.

Conjecture 3.6 (5-flow conjecture)**.** [19, p. 83] *There exists an NZ 5-flow on any bridgeless graph.*

This conjecture had been proved for various classes of snarks. Let us mention some of them. One of the first results concerns graphs, which can be easily adjusted to have an NZ 4-flow.

Theorem 3.7. [7, p. 89] *Let G be a multigraph such that $\Phi(G) = 4$ and $e \in V(G)$ its edge such that $G - e$ is bridgeless. Then, $\Phi(G - e) \leq 5$.*

The previous theorem in fact shows that all snarks of oddness 2 have a NZ 5-flow. However, as for now there is a stronger result.

Theorem 3.8. [14, p. 364] *All cyclically 6-edge-connected snarks of oddness 4 have an NZ 5-flow.*

Moreover, we know some properties of the potential smallest counterexample to the 5-flow conjecture.

Theorem 3.9. [8, p. 142] *If the 5-flow conjecture is false, the smallest counterexample is a cyclically 6-edge-connected snark.*

Theorem 3.10. [9, p. 386] *If the 5-flow conjecture is false, the smallest counterexample has girth 11.*

3.2 Chebyshev flows and flow-pairs

There were some attempts to generalise nowhere-zero flows to real numbers and more dimensions. By generalising integer nowhere-zero flows to real numbers, researchers found out, that restricting only zero flow value is not sufficient, since any flow consisting of non-zero flow values can be made "smaller" and then, the notion of flow number disappears. Therefore, we need to determine a larger interval of real values forbidden as flow values, and hence, one-dimensional real nowhere-zero flows are allowed to use flow values from the interval $[1, r - 1]$. The first attempts of generalisation real flows to more dimensions [13] tried to pick only vectors with the Euclidean norm from the mentioned interval $[1, r - 1]$. It worked, but it did not have such nice properties, as we would expect. However, changing a norm turned out to be a good idea [5], so we will consider only Chebyshev multidimensional nowhere-zero flows.

Definition. Let $G = (V, E)$ be a bridgeless graph, d be a positive integer and $r \geq 2$ be a real constant. A *d-dimensional Chebyshev nowhere-zero r-flow* (or dD ChNZ r-flow) satisfies the conservation constraint (3.1) and the condition $1 \leq \|\varphi(e)\|_\infty \leq r - 1$ for each edge $e \in E$.

The flow number can be defined in a similar way to integral nowhere-zero flows. The only problem is, that the set of real numbers does not need to contain a minimal

element, so formally correct definition would use infimum instead of minimum. However, the infimum of the flow number candidates is always feasible [13, p. 101], so we can define the flow number straightforward with minimum.

Definition. Let G be a bridgeless graph and d be a positive integer. A *d-dimensional Chebyshev flow number* of G is

$$\Phi_d^\infty(G) := \min\{r \mid \exists d\text{D ChNZ } r\text{-flow on } G\}.$$

It has been proved [5, p. 4] that Chebyshev flow number is always rational. Moreover, the authors found some normal form of Chebyshev flows, as stated in the following proposition. Since we need the explicit construction in the latter proofs, we will give a sketch of this proof.

Proposition 3.11. [5, p. 6] *Let G be a graph with $\Phi_d^\infty(G) = p/q$ for some positive integers p, q. Then, there exists a dD ChNZ p/q-flow such that every coordinate of every flow value is an integral multiple of $1/q$.*

Sketch of the proof. If there is a dart with flow value, whose i^{th} coordinate is not an integral multiple of $1/q$, we can find an oriented cycle $\vec{\mathcal{C}}$ containing that dart with a property that no dart of this cycle has flow value with i^{th}, which is an integral multiple of $1/q$. Otherwise, in some vertex, all but one dart have i^{th} coordinate of flow value, which is a multiple of $1/q$, and hence the sum of values on all darts (which is zero by the flow conservation constraint) is not a multiple of $1/q$, a contradiction. Now consider for each $\vec{d} \in \vec{\mathcal{C}}$ the maximal possible adjustment of $\varphi\left(\vec{d}\right)_i$ such that its absolute value remains in the range $[1, p/q - 1]$. Formally,

$$\delta\left(\vec{d}\right) = \begin{cases} p/q - 1 - \varphi\left(\vec{d}\right)_i & \text{if } \varphi\left(\vec{d}\right)_i > 0, \\ -1 - \varphi\left(\vec{d}\right)_i & \text{if } \varphi\left(\vec{d}\right)_i < 0. \end{cases}$$

Clearly $\delta\left(\vec{d}\right) > 0$ for any dart on cycle $\vec{\mathcal{C}}$, since both $p/q - 1, -1$ are integral multiples of $1/q$. Thus, by sending $\delta_{\min} = \min_{\vec{d} \in \vec{\mathcal{C}}} \delta\left(\vec{d}\right)$ along i^{th} coordinate, the number of conflicting coordinates decreases. By repeating this process, we get to a desired normal form. □

Remark. Note that no coordinate exceeds the nearest multiples of $1/q$ during the whole process. As a consequence, when we have a circular NZ r-flow, we can construct an integral NZ $\lceil r \rceil$-flow, where each flow value x is adjusted to either $\lfloor x \rfloor$ or $\lceil x \rceil$.

Now we recall the Seymour's flows from Lemma 3.5 – a 2-flow and a 3-flow with a property, that no edge has both flow values equal to zero. By simply using the value of the 2-flow as the first coordinate and the value of the 3-flow as the second coordinate, we get a Chebyshev nowhere-zero flow. Moreover, this flow gives us a nice bound on the Chebyshev flow number.

Proposition 3.12. [5, p. 8] *Each bridgeless graph has a 2D ChNZ* 3*-flow and hence,* $\Phi_2^\infty(G) \leq 3$.

Again, no graph with $\Phi_2^\infty(G) = 3$ is known. Futhermore, there is no known graph with $\Phi_2^\infty(G) > 5/2$. Therefore, the next conjecture is interesting to examine.

Conjecture 3.13. [5, p. 22] *Each bridgeless graph has a 2D ChNZ* 5/2*-flow and hence,* $\Phi_2^\infty(G) \leq 5/2$.

And similarly as for integral flows, the problem can be reduced to cubic graphs and then to snarks, since 3-edge-colourable cubic graphs have small flow number.

Theorem 3.14. [5, p. 14] *A cubic graph has a 2D ChNZ* 2*-flow if and only if it is* 3*-edge-colourable.*

As an NZ 6-flow and a 2D ChNZ 3-flow can be constructed from two simpler flows, something similar can be made for smaller flow numbers [5, p. 19]. We start by defining this structure called *flow-pair* and then we show that it satisfies desired properties.

Definition. Let $p \leq q$ be the positive integers. A *p/q-flow-pair* of a graph G is a 2-flow φ_2 and a $(p+q+1)$-flow φ_{p+q+1} such that whenever $\varphi_2(e)$ is zero, the value $\varphi_{p+q+1}(e)$ is at least q.

Remark. The structure in the Lemma 3.5 is a 1/1-flow-pair.

One possible ambiguity of this definition is, that since p/q is written as a fraction, the 2/6-flow-pair can be mixed up with the 1/3-flow-pair. However, their existence is equivalent (by a similar argument to Proposition 3.11), so we can WLOG assume that p, q are relatively prime.

Lemma 3.15. [5, p. 19] *A bridgeless graph G with a dp/dq-flow-pair also has a p/q-flow-pair and vice versa.*

We mentioned, that the p/q-flow-pair is a sufficient condition to the existence of some nowhere-zero flows. Those flows are somehow related with the value p/q and that is the reason to write it as a fraction in the definition of the flow-pair.

Proposition 3.16. [5, pp. 19, 20] *Consider a bridgeless graph G with a p/q-flow-pair. Then, there are a 2D ChNZ $(2+p/q)$-flow and a circular NZ $(4+2p/q)$-flow on G.*

Note that if there is a graph, which is a counterexample to Tutte's 5-flow conjecture, it should not have the 1/2-flow-pair. Hence it would be interesting to classify graphs with 1/2-flow-pair. By a computer, it was checked that this flow-pair exists for snarks up to 34 vertices, so it looks like the 1/2-flow-pair exists always.

Conjecture 3.17. [5, p. 22] *Each bridgeless graph has a* 1/2*-flow-pair.*

In general, edges with non-zero value of φ_2 form a set of cycles in G. In cubic graphs, these cycles should be clearly vertex-disjoint. Moreover, if considering other than trivial 1/1-flow-pair, these cycles together cover all vertices.

Lemma 3.18. [5, p. 21] *Let $p < q$ be positive integers. Consider a cubic graph G with a p/q-flow-pair $(\varphi_2, \varphi_{p+q+1})$. Then the set of edges e with a property $\varphi_2(e) \neq 0$ is a 2-factor of G.*

3.3 Rich flows

Recently, in the research of flows, there was an idea to forbid two darts surrounding a vertex to have flow values with the same absolute value [10, 16]. The motivation is, that if we have two darts from the same vertex with $\varphi\left(\overrightarrow{d_1}\right) + \varphi\left(\overrightarrow{d_2}\right) = 0$ (a so-called *confluent pair*), we can split the considered vertex of degree d to vertices of degrees 2 and $d-2$ without influencing flow properties of a graph. On the other hand, a so-called *contrafluent pair*, where $\varphi\left(\overrightarrow{d_1}\right) = \varphi\left(\overrightarrow{d_2}\right)$, requires that in the set of flow values contains some element and its double. However, we cannot find such a pair in a p/q-flow-pair with $p < q$.

Definition. An NZ k-flow is called *rich* if for each pair of adjacent darts d_1, d_2

Proposition 3.19. [16, p. 31] *A cubic graph has a rich NZ 5-flow if and only if it is 3-edge-colourable.*

Conjecture 3.20. [16, p. 33] *There exists a rich NZ 6-flow on any bridgeless cubic graph.*

Proposition 3.21. [16, p. 33] *There exists a rich NZ 11-flow on any bridgeless cubic graph.*

Proposition 3.22. [16, p. 40] *Let G be a bridgeless cubic graph and $e \in V(G)$ its edge such that $R(G-e) = 5$. Then, $R(G) \leq 9$.*

Chapter 4

Cubic multipoles generator

In this chapter, we introduce multipoles as elementary building blocks of graphs. Then we adjust known algorithms for generation of cubic graphs having given girth [1, 2] to obtain a generator of cubic multipoles with given girth.

Definition. A *multipole* is a graph where any edge can have 0, 1 or 2 incident endvertices. Edges with 1 incident endvertex are then called *dangling* and edges with no incident endvertices are called *isolated*.

First we modify an algorithm `snarkhunter` [1].

Definition. A *prime pole* is a multipole which can not be reduced to a smaller one by edge reduction.

Graph K_4^- is a K_4 after deleting any of its edges (so it looks like a diamond).

Lemma 4.1. *A multipole is a prime pole if and only if any of its edges is isolated, dangling, a bridge or has an endvertex in* K_4^-.

Chapter 5

On 1/2-flow-pairs

In this chapter

5.1 Oddness 2

One of the first results on snarks having 5-flow was the class of graphs, that can be constructed from graph with a 4-flow by removing an edge, see Theorem 3.8. We generalise this approach and prove a similar lemma for flow-pairs. This lemma has some additional assumptions needed to finish the proof, so let us discuss, how they change the strength of the proved lemma. At first, the 4-edge-connectivity is not a big deal, since we want to use this lemma mainly for snarks, which are 4-edge-connected. The minimum degree 3 makes no harm, since a vertex of degree 1 cannot be in a bridgeless graph and a vertex of degree 2 can be smoothed without influencing flow properties of graph.

The main issue of this lemma are the two forbidden special cases of flow values surrounding the new edge. Because of them, we cannot state that *"each snark, that can be constructed from graph with a* 0/1*-flow-pair by removing an edge, has a* 1/2*-flow-pair"*. That are the sad news. The good news is that this lemma, similarly as for 1D flows, can be used to prove that snarks with oddness 2 have a 1/2-flow-pair.

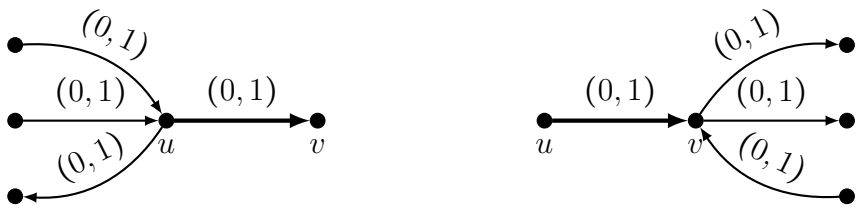
Lemma 5.1. *Let G be a cyclically 4-edge-connected graph with minimum degree at least 3. Consider a pair of vertices $u, v \in V(G)$ and a graph G', which arises from G by adding the edge uv. If G' has a Chebyshev NZ 2-flow φ satisfying*

(a) flow values are integral pairs – $\mathrm{Im}(\varphi) \subseteq \mathbb{Z} \times \mathbb{Z}$,

(b) $\varphi(\overrightarrow{uv}) = (0, 1)$, *and*

(c) these are not the flow configurations around u and v in G',

then G has a 1/2-flow-pair.



Proof. Multiply the second coordinate of φ by two to obtain $\varphi' = (\varphi'_1, \varphi'_2)$. Observe that the flow values of φ' are feasible for a 1/2-flow-pair and $\varphi'(\overrightarrow{uv}) = (0, 2)$. The rest of the proof consists of finding two specific uv-paths in G. Any dart $\overrightarrow{d}$ on this path should satisfy that $\varphi'\left(\overrightarrow{d}\right) + (0, 1)$ is still a feasible flow value of 1/2-flow-pair, because then we can redirect the flow $(0, 2)$ from edge uv to these paths, and then delete the edge uv without violation of the flow conservation constraint.

Let H be an oriented graph such that $V(H) = V(G)$ and

$$E(H) = \{\overrightarrow{xy} \mid xy \in E(G) \wedge (\varphi'_1(\overrightarrow{xy}) \neq 0 \vee \varphi'_2(\overrightarrow{xy}) = 2)\}.$$

One can easily check that this corresponds to the set of darts $\overrightarrow{d}$, for which $\varphi'\left(\overrightarrow{d}\right) + (0, 1)$ is still a feasible flow value. However, some edges in G induce two darts in H. But if we take both of these for some edge e and send $(0, 1)$ through each of them, the resulting flow value on e is the same as before the process. Hence, if we find two dart-disjoint uv-paths in H, the corresponding edges in G' together with the edge uv they form two cycles. After sending $(0, 1)$ along both of these cycles, the flow value at the edge uv becomes zero and all other flow values remain feasible, so we can remove the edge uv and therefore obtain a 1/2-flow-pair of G.

From the oriented version of Menger's Theorem 1.1, there exist two dart-disjoint $\overrightarrow{uv}$-paths in H if and only if each uv-dart-separator of H contains at least 2 darts from the component containing u to the component containing v. So it remains to prove this condition.

Consider any uv-dart-separator $\gamma_{G'}$ of G'. From the generalised flow conservation constraint, $\sum_{\overrightarrow{d} \in \gamma_{G'}} \varphi'\left(\overrightarrow{d}\right) = (0, 0)$. If there is a dart in $\gamma_{G'}$ having a flow value with non-zero first coordinate, there must be at least one more such dart due to the sum of flow values. For both of them there exists a corresponding dart in H. Hence we have at least two darts in the corresponding uv-dart-separator γ_H of H. In the remaining case, only possible flow values on the darts of $\gamma_{G'}$ are $(0, \pm 2)$.

Observe that the amount of darts with flow value $(0, 2)$ must be the same as of darts with flow value $(0, -2)$ due to the zero sum. This also argues that $|\gamma_{G'}|$ is an even number. One of the $(0, 2)$ darts is the new uv edge, and for the others there is a corresponding dart in γ_H. Hence, if $|\gamma_{G'}| > 4$, there are at least three $(0, 2)$ darts in $\gamma_{G'}$, and so at least two $(0, 2)$ darts in γ_H. If $|\gamma_{G'}| = 2$, there should be a vertex of degree 1 in G, which contradicts the assumptions. So we are left with $|\gamma_{G'}| = 4$.

From $|\gamma_{G'}| = 4$ we already know that $|\gamma_G| = 3$. Since G is cyclically 4-edge-connected, at least one of the components separated by γ_G must be an acyclic 3-pole with k inner vertices. Let $d_1 \geq d_2 \geq \cdots \geq d_k \geq 3$ be degrees of its inner vertices and $d_{k+1} = d_{k+2} = d_{k+3} = 1$ be degrees of outer vertices of the dangling edges. We will use that this 3-pole is a tree on $k+3$ vertices, and hence $E(G) = k+2$. Moreover, from the folklore handshaking lemma, we get

$$\begin{aligned} \sum_{i=1}^{k+3} d_i &= 2|E(G)|, \\ 3 + \sum_{i=1}^{k} d_i &= 2k+4, \\ 3k = \sum_{i=1}^{k} 3 \leq \sum_{i=1}^{k} d_i &= 2k+1, \end{aligned}$$

and so $k \leq 1$. This leads to the two forbidden configurations – cut around u and cut around v.

□

Corollary 5.1.1. Each snark G of oddness 2 has a 1/2-flow-pair, and hence, $\Phi_2^\infty(G) \leq 5/2$.

Proof. Consider a 2-factor F with two odd cycles. Then construct an NZ $(\mathbb{Z}_2 \times \mathbb{Z}_2)$-flow φ on G as follows – assign $(1,1)$ to edges in $E \setminus F$ and on the 2-factor F, alternate flow values $(0,1)$ and $(1,0)$. On both odd cycles, there is one conflicting vertex, where the sum of flow values is $(1,1)$. Join these two vertices by a new edge e with an assigned flow value $(1,1)$. Thus, we have an NZ $(\mathbb{Z}_2 \times \mathbb{Z}_2)$-flow φ on $G+e$. It remains to realize, that from Theorem 3.1, both NZ $\mathbb{Z}_2$-flows can be adjusted to NZ 2-flows, and since from the modulo condition their flow values cannot be zero simultaneously, this is a 0/1-flow-pair. Now, from the previous lemma, we get that G has a 1/2-flow-pair. □

5.2 Rich 7-flow

Theorem 5.2. *Let G be a cubic graph with 1/2-flow-pair. Then G has a rich nowhere-zero 7-flow.*

Proof. Denote the 1/2-flow-pair of G as $\varphi = (\varphi_1, \varphi_2)$. First, we construct a circular nowhere-zero 7-flow φ'. Then, we will adjust non-integral values of φ' to φ'', which will be the sought rich nowhere-zero 7-flow.

Let $\varphi'\left(\vec{d}\right) = 9\varphi_1\left(\vec{d}\right)/2 + \varphi_2\left(\vec{d}\right)/2$. We see that

- if $\varphi_1\left(\vec{d}\right) = -1$, then $\varphi'\left(\vec{d}\right) \in \langle -9/2 - 3/2, -9/2 + 3/2 \rangle$,
- if $\varphi_1\left(\vec{d}\right) = 0$, then $\varphi'\left(\vec{d}\right) \in \{-3/2, -2/2, 2/2, 3/2\}$, and

- if $\varphi_1\left(\vec{d}\right) = 1$, then $\varphi'\left(\vec{d}\right) \in \langle 9/2 - 3/2, 9/2 + 3/2\rangle$.

This shows that φ' is a circular nowhere-zero 7-flow.

By algorithm given in the proof of Proposition 3.11, we can construct an integral nowhere-zero 7-flow φ''. Remind that following mentioned algorithm, no flow value exceeds the nearest integers. Hence,

- if $\varphi_1\left(\vec{d}\right) = -1$, then $\varphi''\left(\vec{d}\right) \in \langle -6, -3\rangle$,
- if $\varphi_1\left(\vec{d}\right) = 0$, then $\varphi''\left(\vec{d}\right) \in \{-2, -1, 1, 2\}$, and
- if $\varphi_1\left(\vec{d}\right) = 1$, then $\varphi''\left(\vec{d}\right) \in \langle 3, 6\rangle$.

However, this is the key observation to prove that φ'' is rich. For the sake of contradiction, assume a vertex v in which flow values of φ'' in two outgoing darts are the same, denote it by x. Hence, the third flow value is $2x$, which is in absolute value greater. Hence, the two equal flow values must have the smallest absolute value. From Lemma 3.18, each vertex is surrounded by one dart with $\varphi_1\left(\vec{d}\right) = 0$ and two darts with $\varphi_1\left(\vec{d}\right) = 1$. But absolute values of $\varphi''\left(\vec{d}\right)$ on darts with $\varphi_1\left(\vec{d}\right) = 0$ are strictly smaller than absolute values of $\varphi''\left(\vec{d}\right)$ on darts with $\varphi_1\left(\vec{d}\right) \neq 0$. Therefore, there cannot be two darts with the same smallest absolute value, a contradiction. In final, we get that φ'' is a rich NZ 7-flow.

□

Chapter 6

Chebyshev flow number of snarks

6.1 Chebyshev flow number of generalised Blanuša snarks

6.2 General lower bound for Chebyshev flow number

In one-dimensional flows, there has been proved a lower bound on the flow number of a snark

$$\Phi_1(G) \geq 4 + 1 \Big/ \left\lceil \frac{n-4}{8} \right\rceil$$

depending on the number n of its vertices [11, p. 14]. Moreover, there is also a meaningful lower bound on the two-dimensional Chebyshev flow number of the Petersen graph P [18, p. 99]. It states that $\Phi_2^\infty(P) \geq 5/2$. We provide generalisation of this bound for any snark. Additionally, the bounding number is roughly a half of the bounding number in one dimension.

Proposition 6.1. [5, pp. 15, 16] *Let G denote a snark of order n. Then $\Phi_2^\infty(G) \geq 2 + 1/\xi$, where $\xi = \left\lfloor \frac{n-2}{4} \right\rfloor$.*

Proof. Assume by contradiction that there exists a 2-dimensional flow φ of G such that $\varphi = (\varphi_1(e), \varphi_2(e))$ for each edge $e \in E(G)$, with $\|\varphi(e)\|_\infty \geq 1$ and $\varphi_i(e) \in (-1 - 1/\xi, 1 + 1/\xi)$ for $i = 1, 2$.

We say an edge $e \in E(G)$ to be *large* with respect to φ_i if $|\varphi_i(e)| \in [1, 1 + 1/\xi)$, *small* otherwise. Observe that an edge e can be large with respect to both φ_1 and φ_2, but it cannot be small with respect to both φ_1 and φ_2, for otherwise $\|\varphi(e)\|_\infty < 1$.

Denote by S_i the subgraph of G induced by the small edges with respect to φ_i and by L_i the one induced by the large edges with respect to φ_i, $i = 1, 2$. By previous

observation at least one between S_1 and S_2 has at most $\lfloor |E(G)|/2 \rfloor = \lfloor \frac{3n}{4} \rfloor$ edges, say S_1.

Claim. S_i is a spanning subgraph of G and $\Delta(S_i) \leq 2$, for $i = 1, 2$.

Proof. is this needed?? Observe that $\Delta(L_i) \leq 2$, because the sum of three real numbers all with absolute value in the interval $[1, 1 + 1/\xi)$ cannot give 0 as a result, making the Kirkoff's law impossible to be satisfied by φ around a vertex of G. Hence S_i is spanning, for otherwise $\Delta(L_i) = 3$ and $\Delta(S_i) \leq 2$, for otherwise $\Delta(L_{3-i}) = 3$. □

Claim. If $C \subseteq E(L_i)$ is an odd edge-cut of G for $i = 1, 2$, then $|C| \geq 2\xi + 3$.

Proof. Consider an odd edge-cut C of G containing $2k + 1$ large edges, separating components G_1, G_2. WLOG assume $\varphi_i(e) \geq 0$ for every edge e of C and more edges are directed towards G_2. Then, there are at least $k+1$ edges towards G_2, resulting in total inflow at least $(k+1) \cdot 1$. Analogously, the total outflow is strictly less than $k \cdot (1+1/\xi)$. Together with the Kirkoff's law, this leads to $k + 1 < k \cdot (1 + 1/\xi)$, which is equivalent to $k > \xi$. □

Claim. A path of length $2k$ cannot be a connected component of S_i, for $k = 1, 2, \ldots, \xi - 1$ and $i = 1, 2$.

Proof. For the sake of contradiction assume there is a path of a length $2k$, $k < \xi$ in S_i. Then the edges adjacent to this path are in L_i and they form an odd edge-cut of G, containing at most $2k + 3 < 2\xi + 3$ edges, which is in contradiction with the Claim 2. □

Claim. A cycle of length $2k+1$ cannot be a connected component of S_i, for $k = 1, 2, \ldots, \xi$ and $i = 1, 2$.

Proof. For the sake of contradiction assume there is a cycle of a length $2k + 1$, $k < \xi + 1$ in S_i. Then the edges adjacent to this cycle are in L_i and they form an odd edge-cut of G, containing at most $2k + 1 < 2\xi + 3$ edges, which is in contradiction with the Claim 2. □

Claim. $E(S_i)$ cannot contain a perfect matching of G, for $i = 1, 2$.

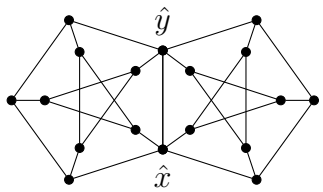
Proof. For the sake of contradiction assume that $E(S_i)$ contains a perfect matching M of G. Then also $E(L_{3-i})$ contains a perfect matching M of G. Next, $F = E(G) \setminus M$ is a 2-factor of G. Note that F must contain an odd cycle of length $2k + 1 \leq \frac{n}{2}$ explain more – here's implicitly used that G is snark. This is equivalent with $k \leq \xi$. Then the edges adjacent to this cycle are in L_{3-i} and they form an odd edge-cut of G, containing at most $2k + 1 \leq 2\xi + 1$ edges, which is in contradiction with the Claim 2. □

Note that by the Claim 3, each even path in S_1 contains at least 2ξ edges. Similarly by the Claim 4, each odd cycle in S_1 contains at least $2\xi+3$ edges. Let *odd components* denote odd cycles and even paths. Since $4\xi = 4\lfloor\frac{n-2}{4}\rfloor > \lfloor\frac{3n}{4}\rfloor \geq E(S_1)$ obviously holds for any $n \geq 10$, S_1 may contain at most one odd component. On the other hand, S_1 contains even number of odd componentss. As a result, S_1 contains only odd paths and even cycles, but then also a perfect matching of G, which is in contradiction with the Claim 5. □

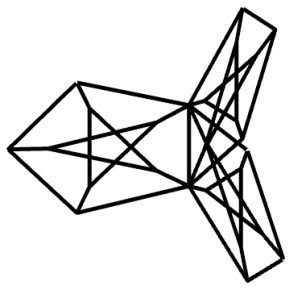
For now, we are aware of seven snarks meeting this bound [5, p. 17] – Petersen graph P with $\Phi_2^\infty(P) = 5/2$, both Blanuša snarks $B^{(1)}, B^{(2)}$ with $\Phi_2^\infty(P) = 9/4$ and 4 out of 6 snarks on 20 vertices with 2D Chebyshev flow number also equal to 9/4. From computational results, we know that no other example of tightness exists on at most 26 vertices [5, p. 17]. Although we do not know about any other snark meeting this bound, we think, the bound is not so far from reality, since we found an infinite class of graphs, that are nearly cubic and meet this bound.

The mentioned class was already introduced when constructing graphs with given circular flow number [15, p. 312].

Definition. Let Q be the Petersen graph without an edge (since the Petersen graph is edge-transitive, it does not matter, which one is removed) and denote its two vertices of degree two by x and y. Then, a *Petersen fan* PF_n is a graph, which arises from n copies of Q by identifying all vertices corresponding to x in a vertex $\hat{x}$, similarly all vertices corresponding to y in $\hat{y}$ and finally joining $\hat{x}$ and $\hat{y}$ by an edge.



(a) $n = 2$



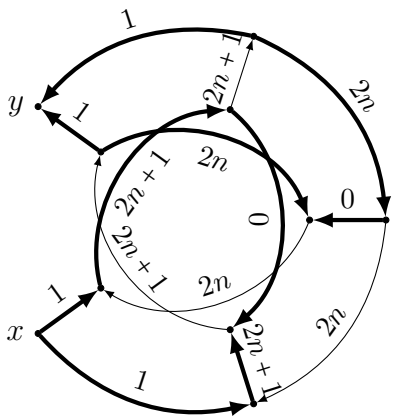
(b) $n = 3$

Figure 6.1: Two small examples of PF_n graphs

Lemma 6.2. [15, p. 312] *For each positive integer n, $\Phi_1(PF_n) = 4 + 1/n$.*

From 3.16, one can easily see, that if PF_n admits a t-flow-pair, then $t \geq 1/(2n)$. And for $t = 1/(2n)$, the flow-pair also exists.

Proposition 6.3. *For each positive integer n, PF_n admits a $(1/(2n))$-flow-pair.*



Proof. We begin the proof by defining flow values of flow-pair (φ_1, φ_2) on one copy of Q. The edges of 2-factor induced by φ_1 are bolded. We can check that in each vertex except x and y, flow conservation costraint is satisfied and non-bolded edges use only flow values $2n$ and $2n+1$. Hence, all flow values used in one copy of Q are feasible. Moreover, when we join n copies of Q, the sum of flow values outgoing from $\hat{x}$ will be $(0, 2n)$, and the same will be the sum of flow values ingoing to $\hat{y}$. However, this is also a feasible flow value, so by setting $\varphi\left(\overrightarrow{\hat{y}\hat{x}}\right) = (0, 2n)$, we get the desired $1/(2n)$-flow-pair of PF_n. □

This Proposition also gives an upper bound on Chebyshev flow number of PF_n.

Corollary 6.3.1. For each positive integer n, $\Phi_2^\infty(PF_n) \leq 2 + 1/(2n)$.

By a computer, we verified that $\Phi_2^\infty(PF_3) = 2 + 1/6$ and $\Phi_2^\infty(PF_4) = 2 + 1/8$, so we expect that the inequality can be replaced by the equality.

Conjecture 6.4. *For each positive integer n, $\Phi_2^\infty(PF_n) = 2 + 1/(2n)$.*

Note that PF_n is of order $8n+2$. Even though PF_n is not cubic for $n \geq 2$ (due to two vertices of degree $2n+1$), observe that by inserting the order into the bound from Proposition 6.1, we get $1/(2n)$. Therefore, if we were able to split each of the two vertices of high order into vertices of degree 2 and one vertex of degree 3 without increasing the flow number or creating a 3-edge-colourable graph, after smoothing vertices of degree 2 we will obtain a graph meeting the lower bound.

This is possible for $k = 2$, where we can obtain both Blanuša snarks [15, p. 316]. For $k = 3$ we already know this is not possible, because all snarks on 26 vertices have 2D Chebyshev flow number at least $2 + 1/4$. However, we are able to reduce the degrees of these vertices so that in the resulting graph, only one vertex is not of degree 3, and its degree is 5. Then we replaced this vertex by a 5-cycle, and since the flow number still was the same, we constructed a snark on 30 vertices with 2D Chebyshev flow number $2 + 1/6$. Hence, we suspect, no snark with 2D Chebyshev flow number $2 + 1/6$ on 28 vertices exists.

In summary, adjusting PF_n without changing the flow number can lead to small cubic graphs with small flow number. Now we will show one general construction, giving an upper bound on order of graph with sufficiently small flow number. This method is inspired by the similar construction for circular flow number [12, pp. 193-195].

Proposition 6.5. *Let G be a graph with p/q-flow-pair. Then there exists a cubic graph G' with p/q-flow-pair.*

Proof. Fix a p/q-flow-pair of G and consider a vertex v with degree d greater than 3. We aim to construct a cubic d-pole, which will be the replacement of v in G. Therefore, the multiset of the flow values on dangling edges of this pole should be same as the multiset of flow values surrounding v in the original graph.

Let us start with the d dangling edges with corresponding flow values. Now, we attach graphs to the dangling edges, to obtain only dangling edges with flow values $(0, \pm(p+q)), (\pm 1, 0)$ in the end. To get rid of unwanted edges with zero first coordinate and $a > 0$, we will use the graph in Figure 6.2. The graph for $a < 0$ can be obtained in a similar manner.

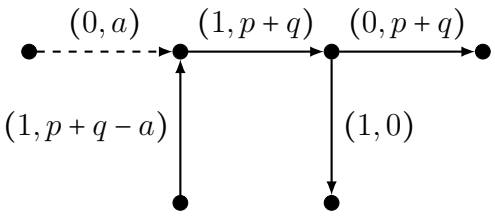


Figure 6.2: TODO

So now we are left with the unwanted edges with non-zero first coordinate. Take two dangling edges with first coordinates equal to 1 and second coordinates $-p-q < a \leq b < p+q$. Let $x = \min\{p+q-b, p+q+a\}$. We attach $\lfloor x/p \rfloor$ copies of the graph in Figure 6.3 with $\delta = p$ and then another copy with $\delta = x \mod p$. This causes that we get either $(1, -p-q)$ or $(1, p+q)$ on one of the dangling edges. To this dangling edge, we can attach dangling edges with flow values $(1, 0)$ and $(0, \pm(p+q))$, and hence reduce the number of unwanted edges. Since this can be done while there are at least two dangling edges with the first coordinate equal to 1, we will end up with at most one conflicting edge, whose first coordinate of flow value is 1. By a similar reasoning, we can deal with dangling edges, where first coordinate of flow value is -1.

At this moment, the majority of dangling edges have flow values $(\pm 1, 0)$ and $(0, \pm(p+q))$. The only two possible exceptions are $(1, s)$ and $(-1, t)$ for some $-p-q < s, t < p+q$. From the flow conservation constraint, we deduce that $s+t \equiv 0 \pmod{p+q}$. If $s+t = 0$, we can attach those dangling edges and smooth their common vertex of degree 2. On

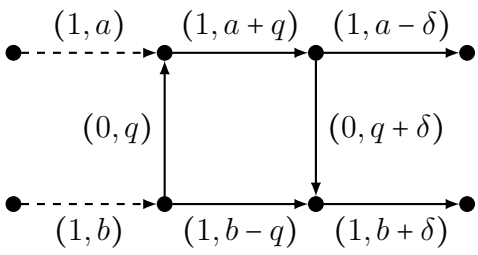


Figure 6.3: TODO

the other hand, if $s + t = \pm(p + q)$, we will additionally attach a $(0, \pm(p + q))$ dangling edge to fulfill the flow conservation constraint. As the last step, we realize, that from the flow conservation constraint, the number of $(1, 0)$ dangling edges is equal to the number of $(-1, 0)$ dangling edges (and same for the $(0, \pm(p + q))$ edges), so we can attach edges with opposite flow values and smooth their common vertex. Finally, we ended up with (not necessarilly connected) cubic d-pole with desired properties, and hence, we constructed a cubic graph with p/q-flow-pair. □

Corollary 6.5.1. For any $r > 2$, there exists a non-colourable cubic graph, whose 2D Chebyshev flow number is at most r.

6.3 Three-dimensional Manhattan flow number

Proposition 6.6. *For any bridgeless graph G it holds $\Phi_3^1(G) \leq \frac{5}{2}$.*

Proof. Define the following vectors in $\mathbb{R}^3$:

$$q_1 = \left(\frac{3}{8}, \frac{3}{8}, -\frac{1}{4}\right),$$
$$q_2 = \left(\frac{3}{8}, -\frac{1}{4}, \frac{3}{8}\right),$$
$$q_3 = \left(-\frac{1}{4}, \frac{3}{8}, \frac{3}{8}\right).$$

Note that since the vectors are cyclic, it suffices to find norms of $q_1, q_1 + q_2, q_1 - q_2, q_1 + q_2 + q_3$ and $q_1 + q_2 - q_3$. But

$$\|q_1\|_1 = 1,$$
$$\|q_1 + q_2\|_1 = 1,$$
$$\|q_1 - q_2\|_1 = \frac{5}{4},$$
$$\|q_1 + q_2 + q_3\|_1 = \frac{3}{2},$$
$$\|q_1 + q_2 - q_3\|_1 = \frac{3}{2}.$$

By the 8-flow theorem, the graph G can be covered by three cycles C_1, C_2, C_3. We orient each cycle arbitrarily and assign to each oriented cycle C_i the flow vector q_i. The total flow is then defined as the sum of these three cycle flows.

By construction and the norm evaluations above, for every edge $e \in E$, the resulting flow satisfies

$$\|\varphi(e)\|_1 \in \left\{1, \frac{5}{4}, \frac{3}{2}\right\}.$$

$\square$

Conclusion

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